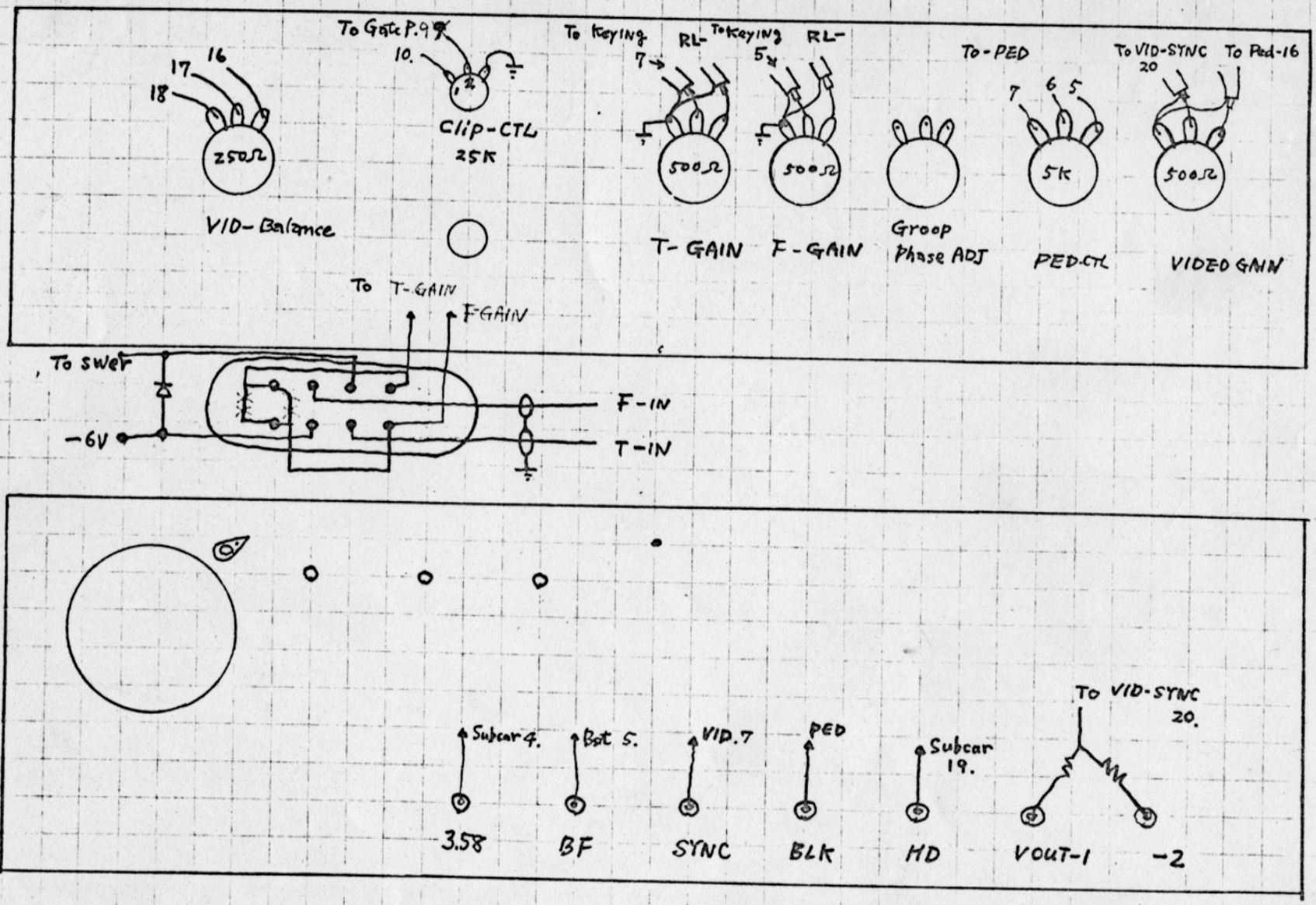
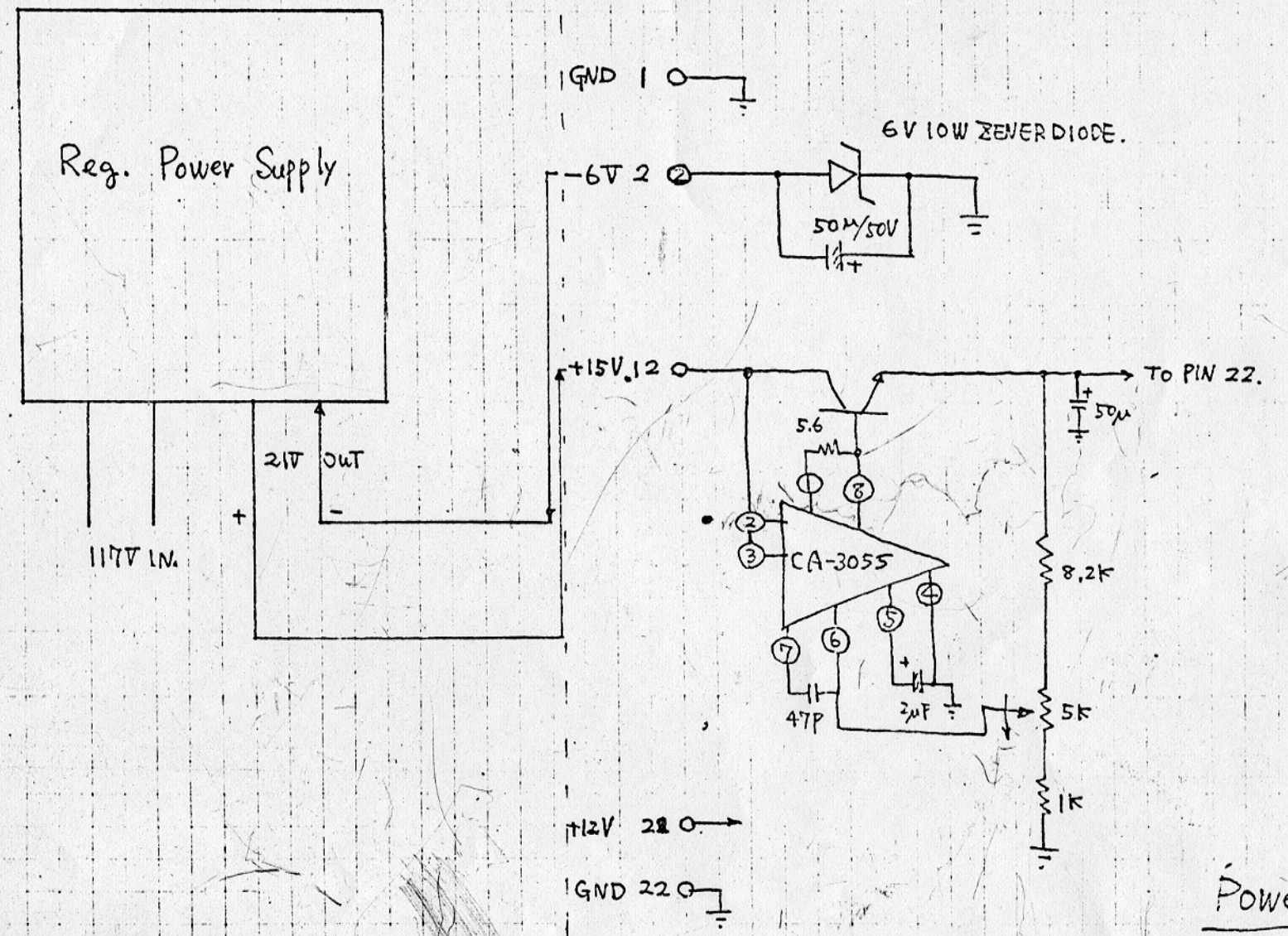
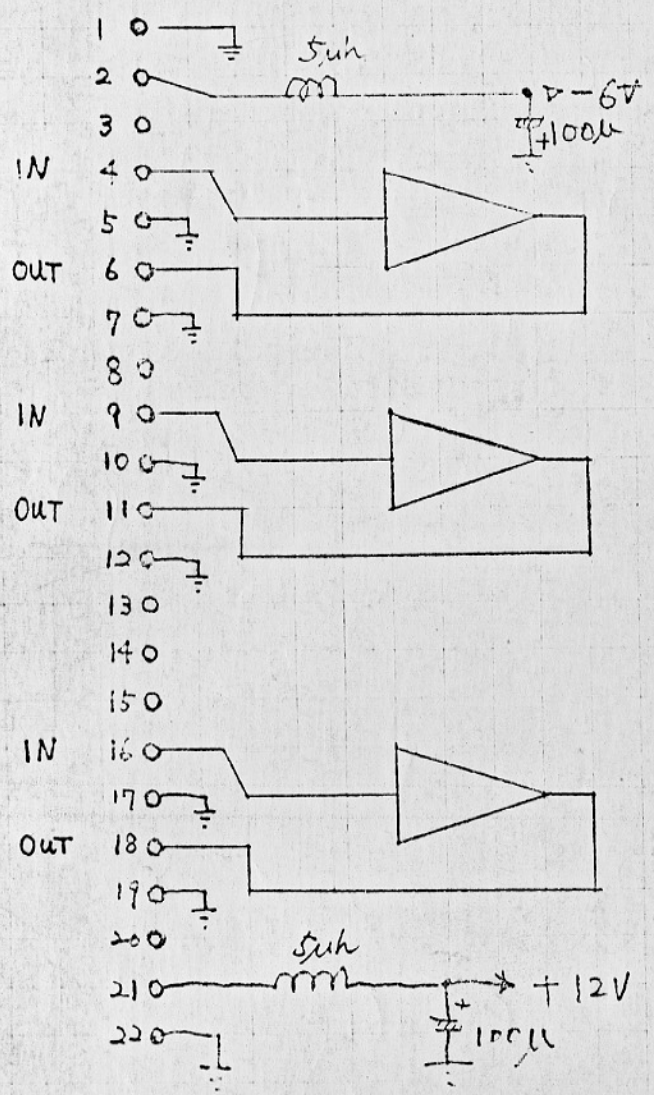
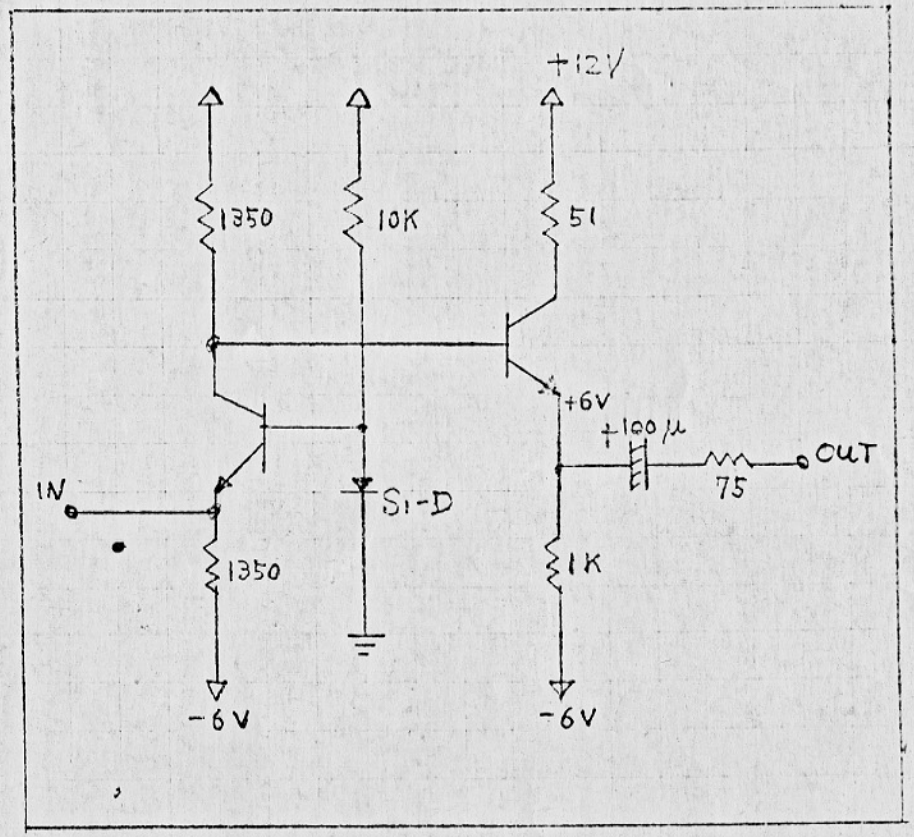

To Gate P.9
10.
Clip-CTL
25K
To keying
RL-
7
To keying
RL-
5
To-PED
To VID-SYNC
20
To Ped-16
18
17
16
250Ω
VID-Balance
500Ω
T-GAIN
500Ω
F-GAIN
Group
Phase ADJ
7
6
5
5K
PED-CTL
500Ω
VIDEO GAIN
To T-GAIN
F-GAIN
To swer
-6V
F-IN
T-IN
Subcar 4.
Bst 5.
VID.7
PED
Subcar
19.
To VID-SYNC
20.
3.58
BF
SYNC
BLK
HD
VOUT-1
-2

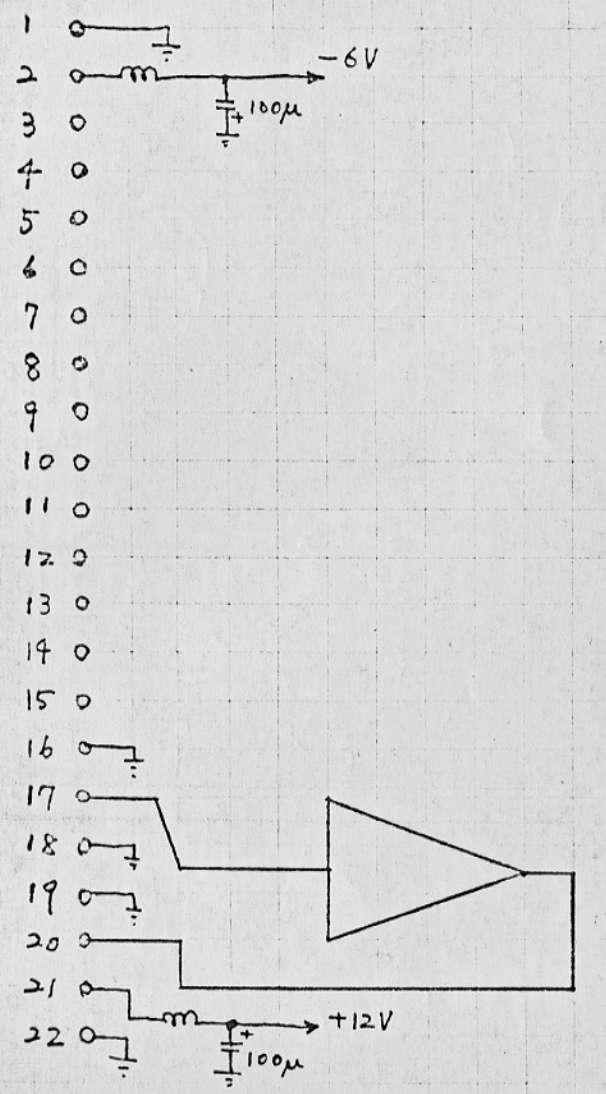


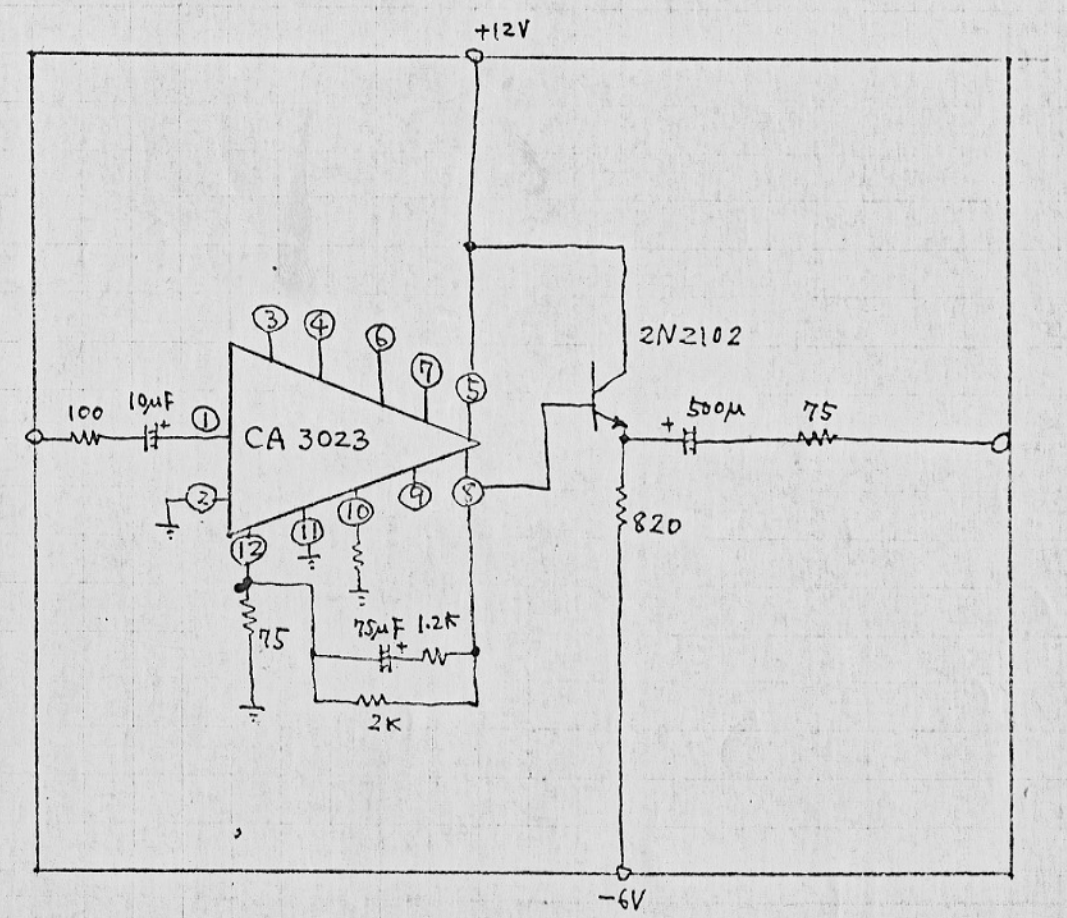
Power Supply


1
2
3
IN 4
5
OUT 6
7
8
IN 9
10
OUT 11
12
13
14
15
IN 16
17
OUT 18
19
20
21
22
5uh
-6V
100µ
5uh
+12V
100µ

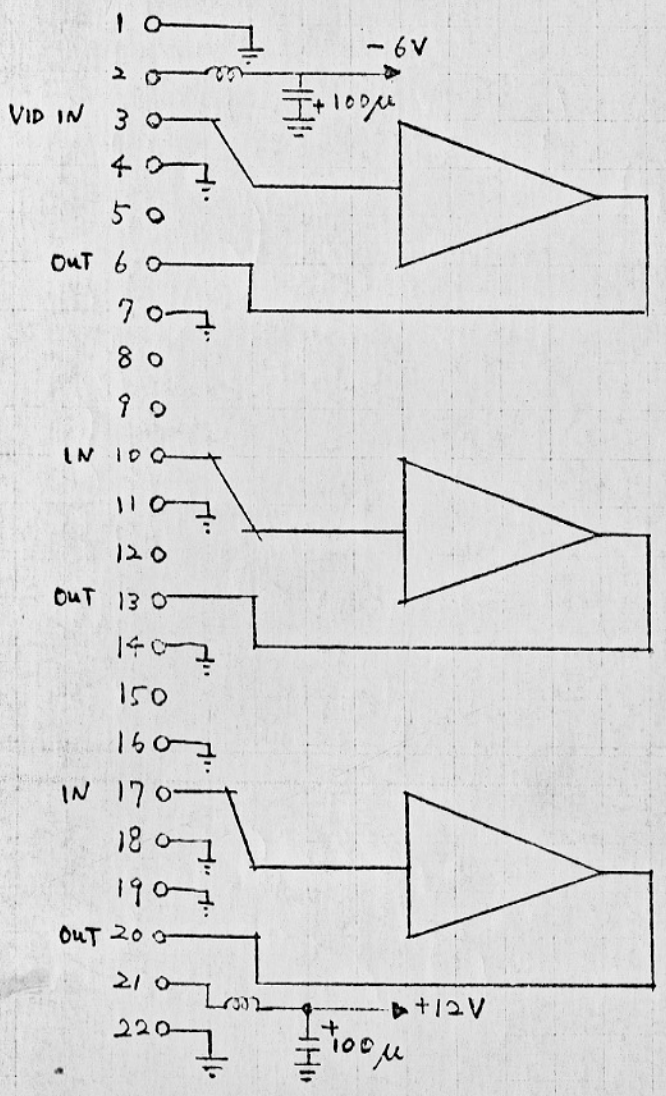

+12V
1350
10K
51
+6V
100µ
OUT
75
IN
SI-D
1350
1K
-6V
-6V

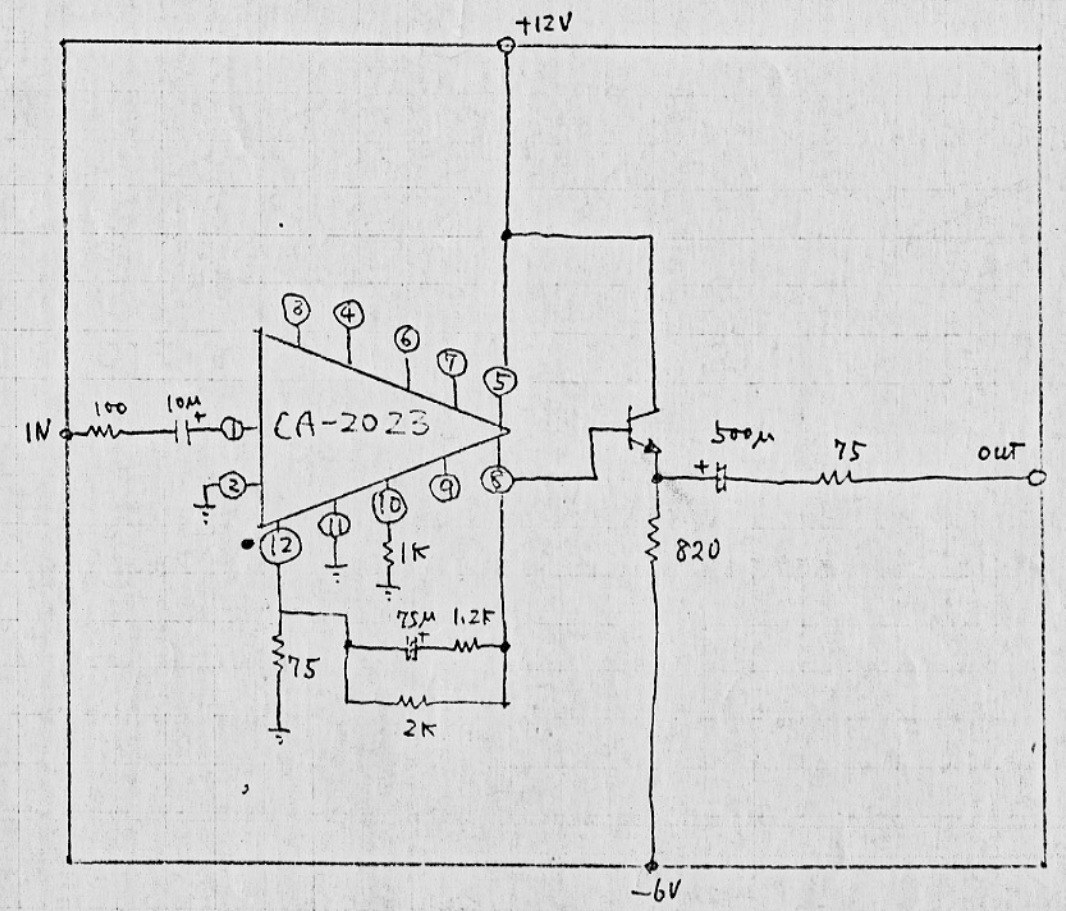
Mixing Amp


1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
-6V
100μ
+12V
100μ

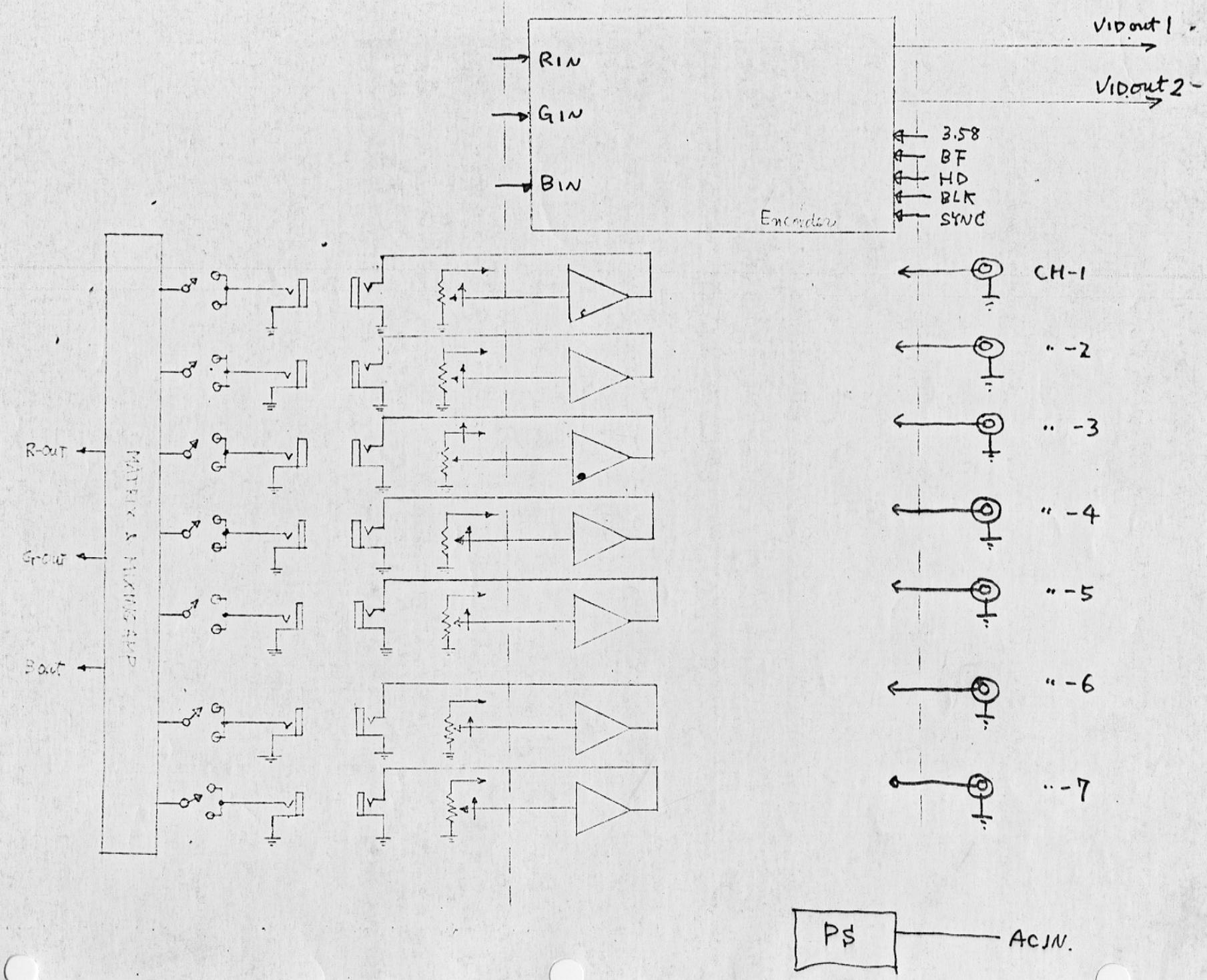

+12V
2N2102
100
10μF
CA 3023
500μ
75
820
75
75μF
1.2K
2K
-6V

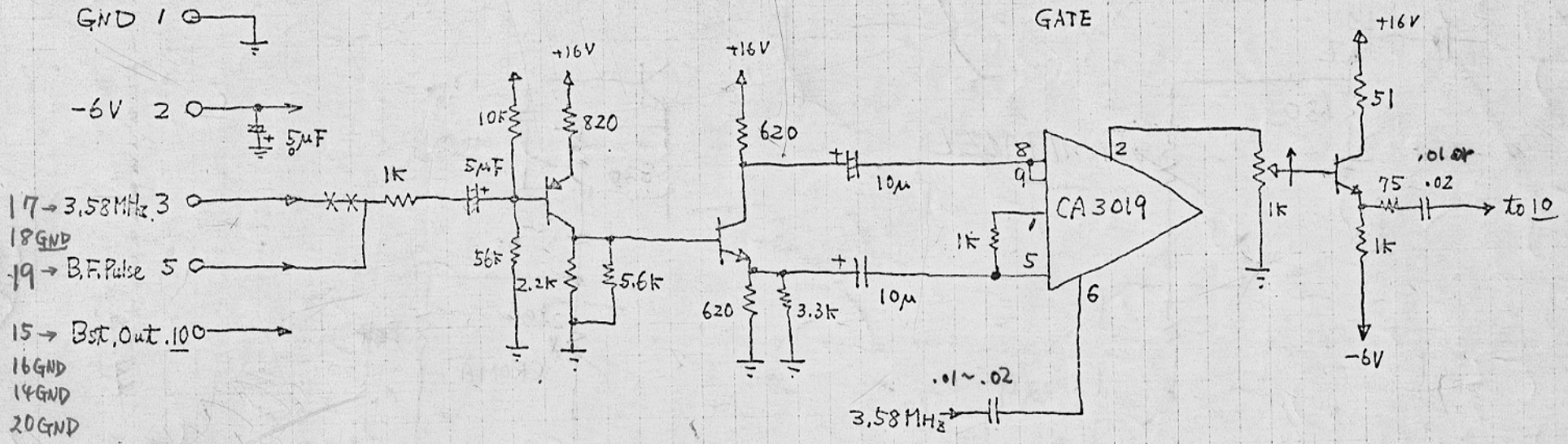
Amp.

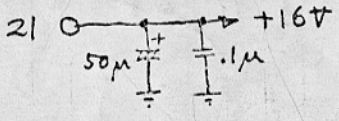

1
2
VID IN 3
4
5
OUT 6
7
8
9
IN 10
11
12
OUT 13
14
15
16
IN 17
18
19
OUT 20
21
22
-6V
+100μ
+12V
+100μ


+12V
IN
100
10M
CA-2023
500μ
75
OUT
820
1K
75
75μ 1.2K
2K
-6V

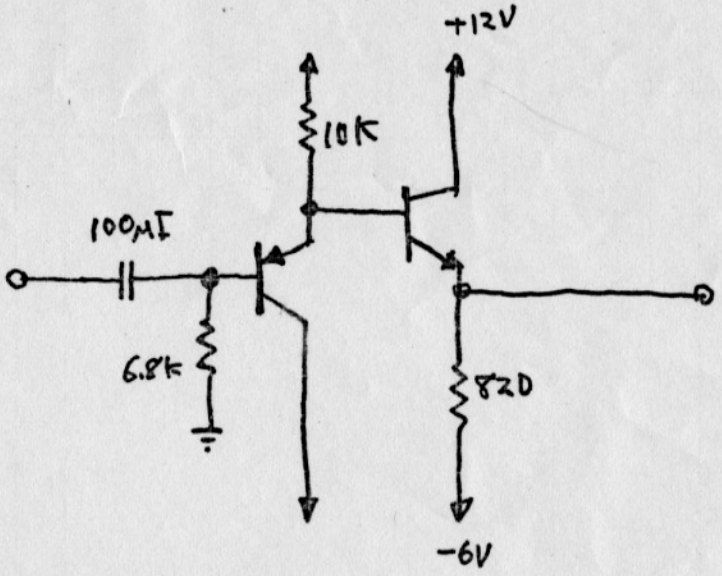
Amp.


RIN
GIN
BIN
Encoder
VIDout 1
VIDout 2
3.58
BF
HD
BLK
SYNC
MATRIX & MIXING AMP
R-OUT
G-OUT
B-OUT
CH-1
" -2
" -3
" -4
" -5
" -6
" -7
PS
ACIN.





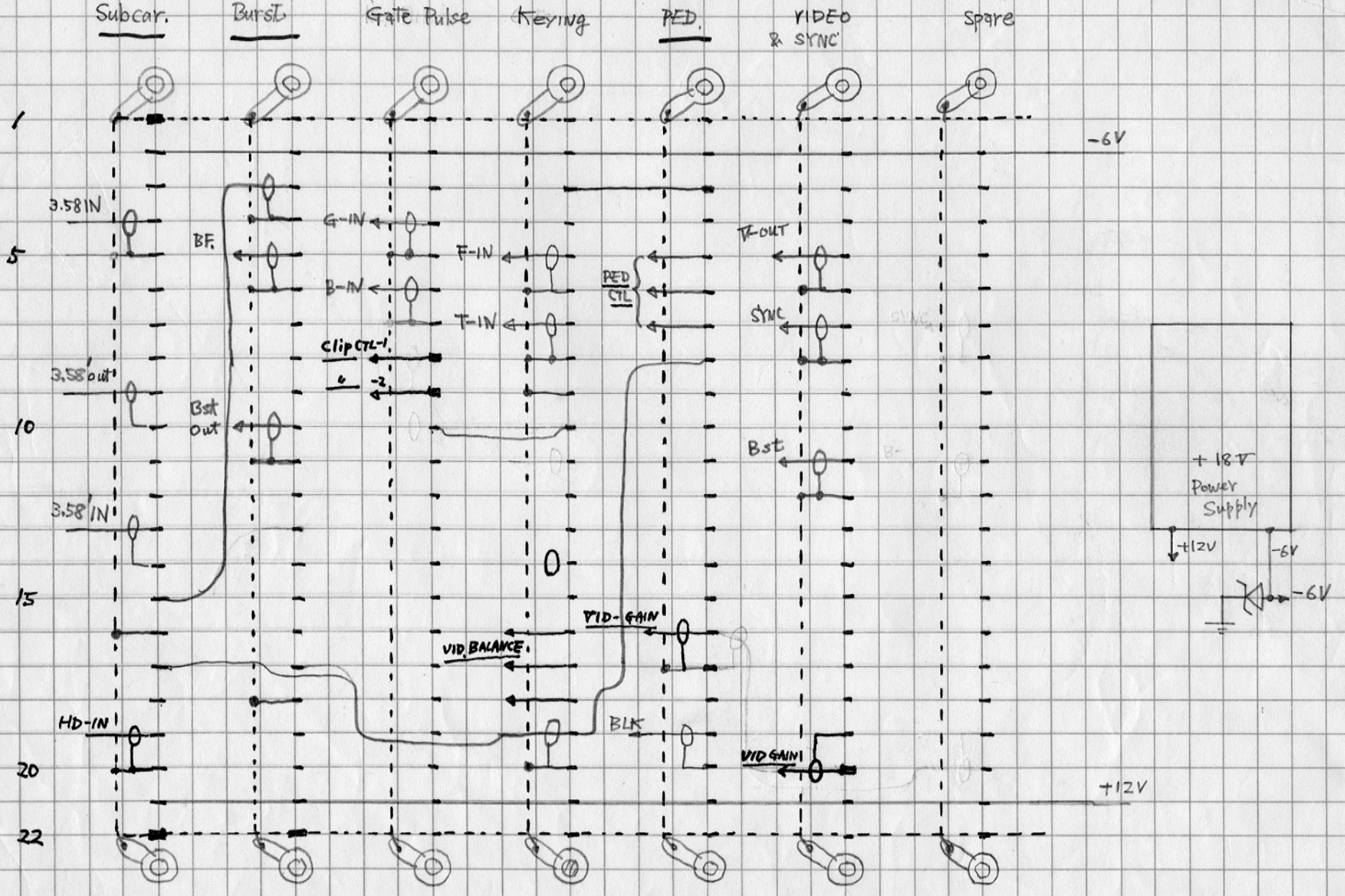
Burst Generator

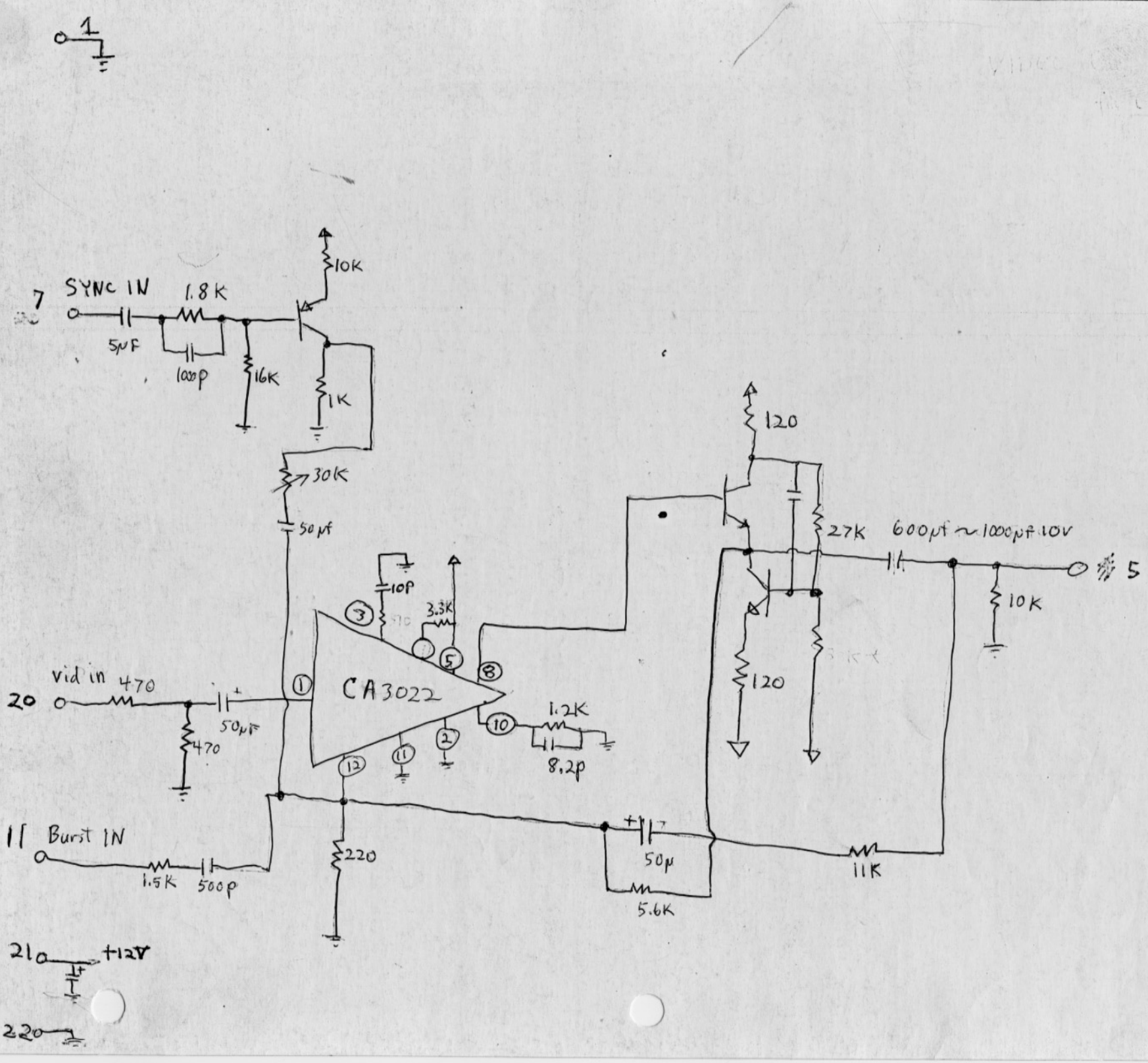


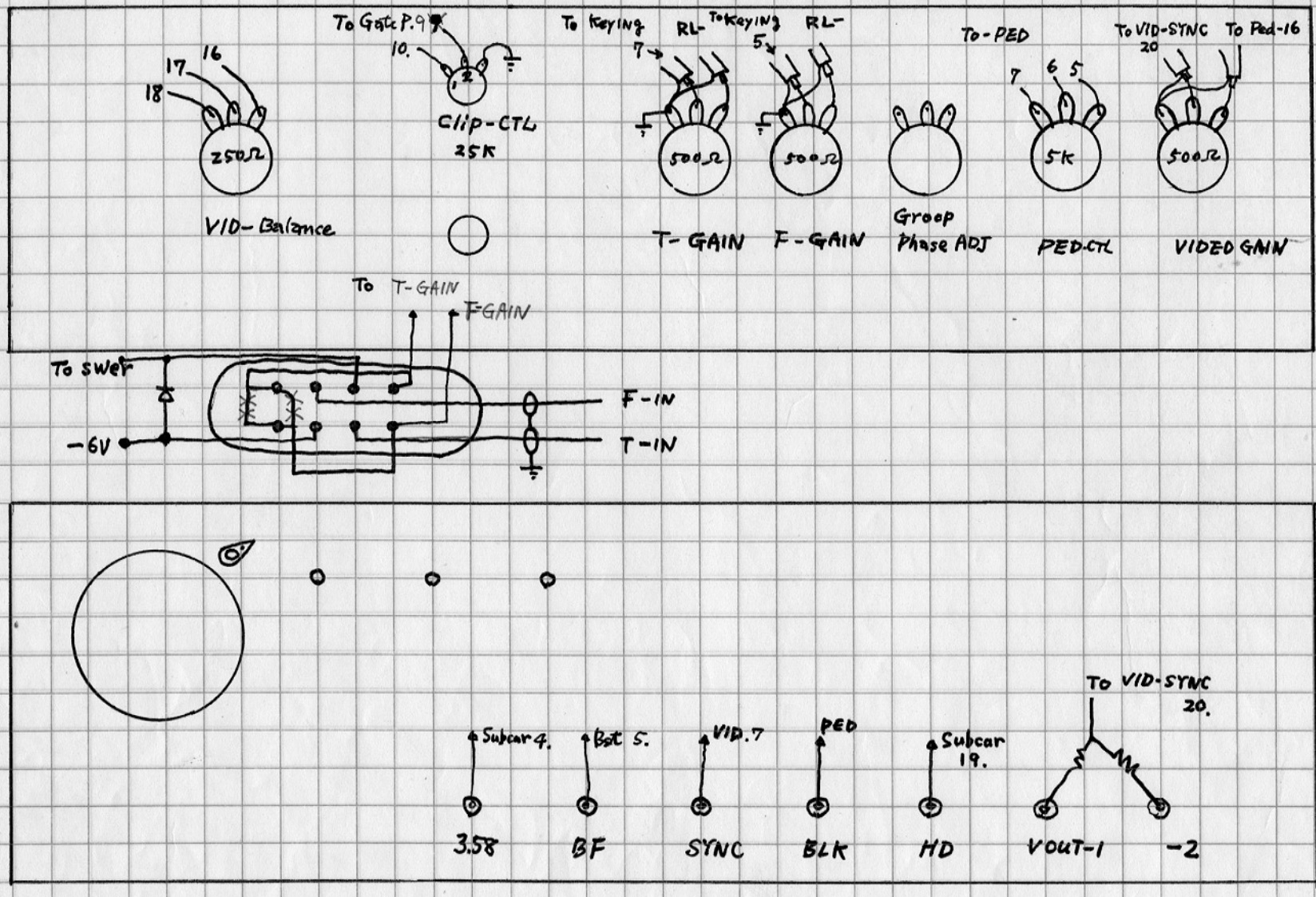
SAME AS Above

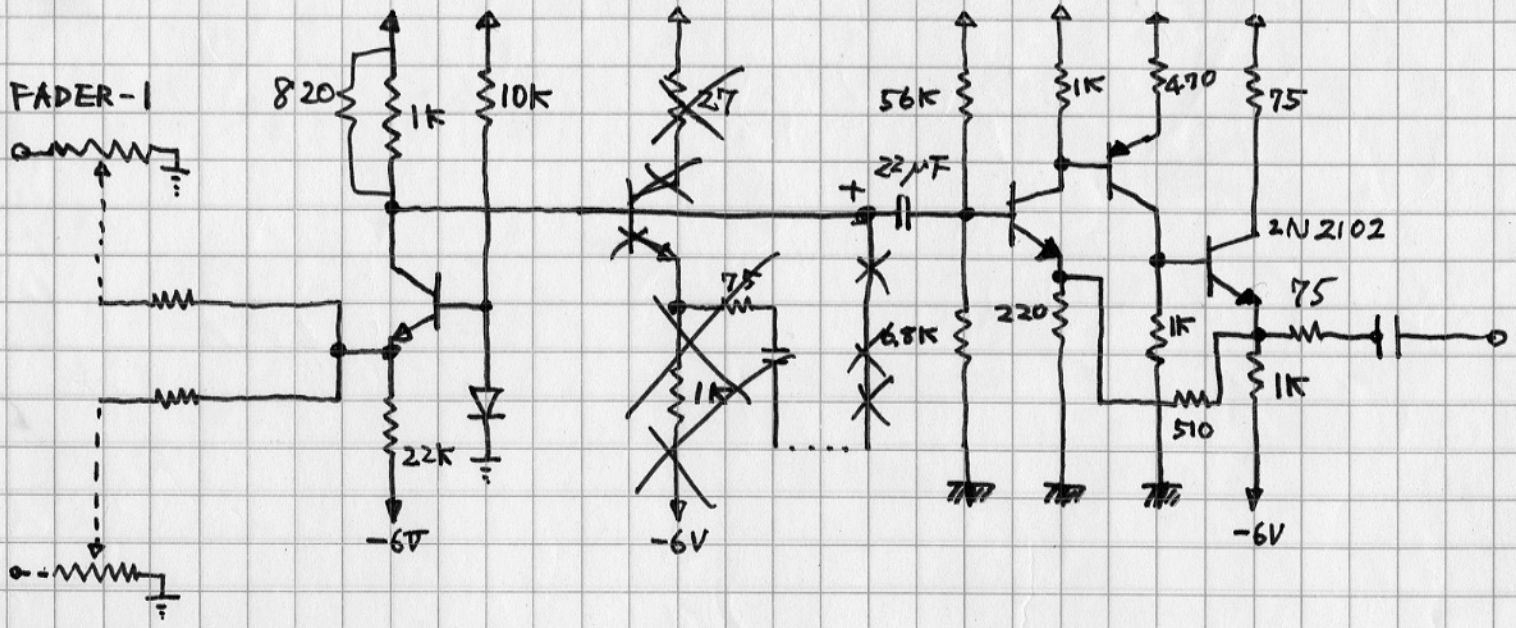
SAME AS Above

SWITCHER INPUT
EMITTER
Follower


Subcar.
Burst
Gate Pulse
Keying
PED.
VIDEO & SYNC
Spare
1
5
10
15
20
22
-6V
+12V
3.58 IN
3.58 out
3.58 IN
HD-IN
BF.
Bst out
G-IN
B-IN
Clip CTL-1.
4
-2
F-IN
T-IN
PED CTL
VID. BALANCE.
VID. GAIN
BLK
T-OUT
SYNC
Bst
VID GAIN
+18V Power Supply
+12V
-6V
-6V


1
7
SYNC IN
1.8K
10K
5μF
1000p
16K
1K
30K
50μf
10p
3.3K
3
5
8
vid in
470
20
1
CA3022
50μF
470
10
1.2K
8.2p
12
11
2
11
Burst IN
1.5K
500p
220
50μ
5.6K
11K
120
27K
600μf ~ 1000μf 10V
5
10K
120
21
+12V
220

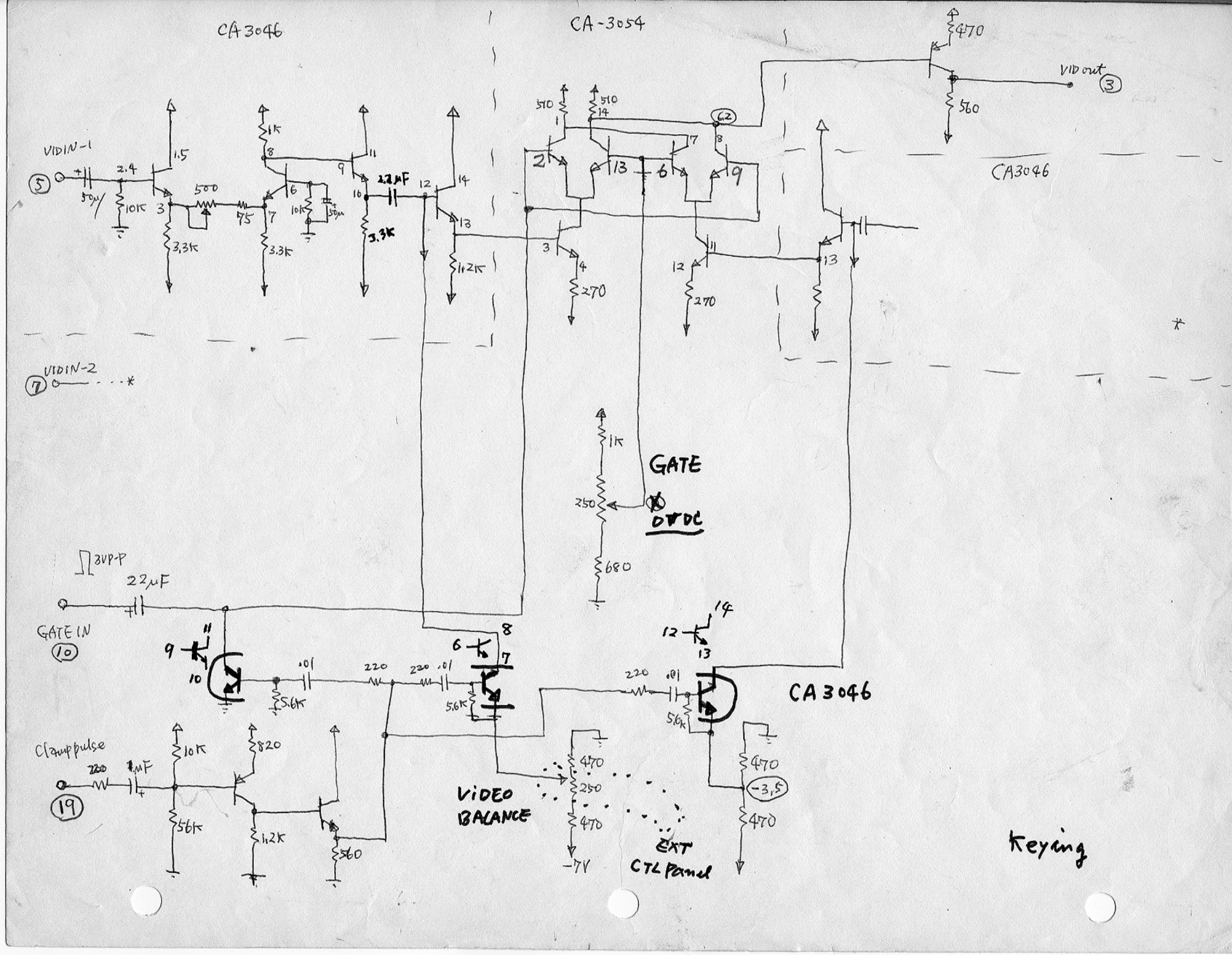

To Gate P.9
10.
Clip-CTL
25K
To Keying
7
RL-
To Keying
5
RL-
To-PED
To VID-SYNC
20
To Ped-16
17
16
18
250Ω
500Ω
500Ω
7
6
5
5K
500Ω
VID-Balance
T-GAIN
F-GAIN
Group
Phase ADJ
PED-CTL
VIDEO GAIN
To T-GAIN
F-GAIN
To SWEP
-6V
F-IN
T-IN
Subcar 4.
Bst 5.
VID.7
PED
Subcar
19.
To VID-SYNC
20.
3.58
BF
SYNC
BLK
HD
VOUT-1
-2

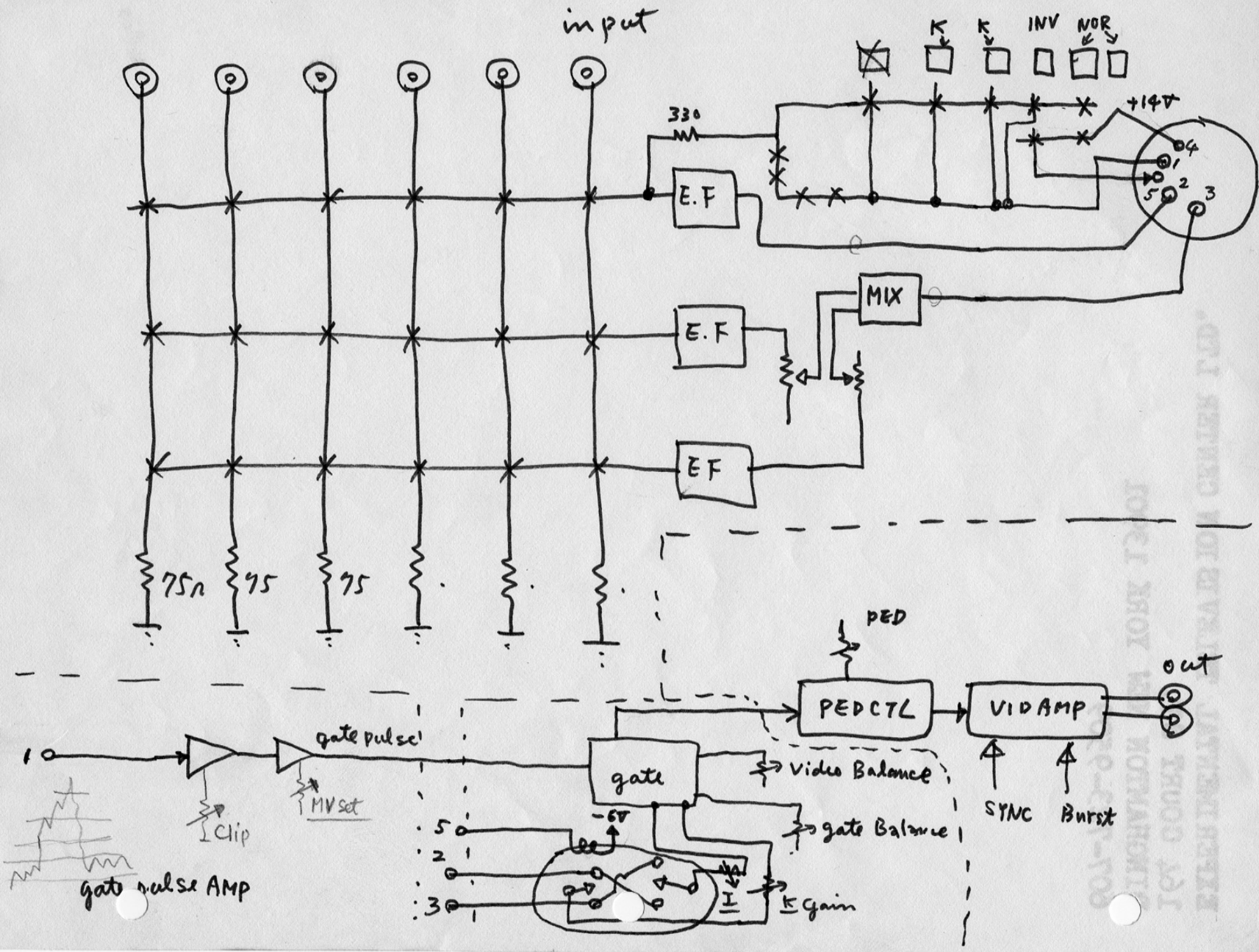


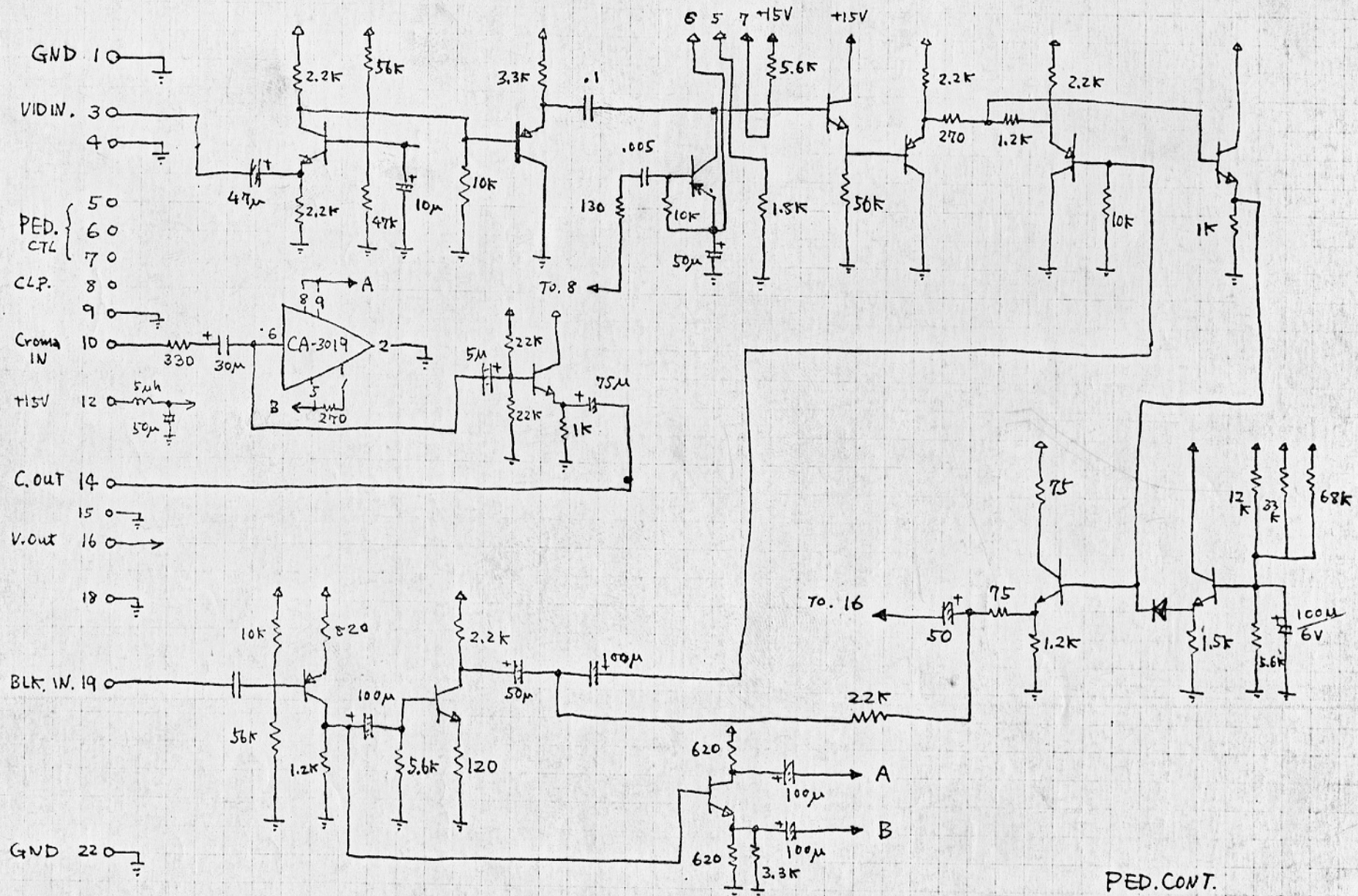
PNP ... 2N4402 or Equivalent

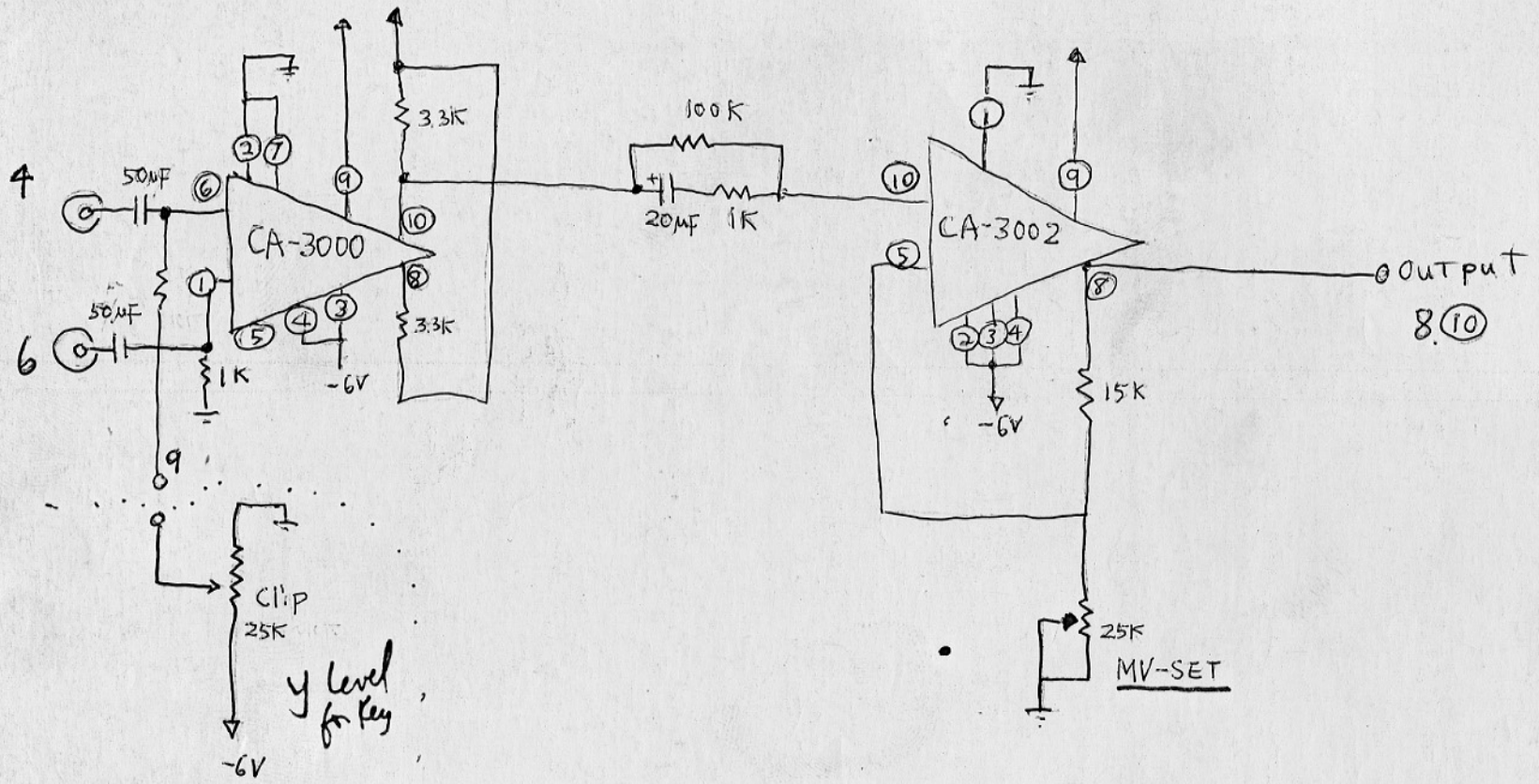
NPN ... 2N4400 " "

MIXER - FADER OUTPUT AMP


CA3046
CA-3054
VIDIN-1
VIDIN-2
VID out
CA3046
GATE
DVDC
GATE IN
22μF
3VP-P
Clamppulse
VIDEO BALANCE
EXT CTL Panel
CA3046
-7V
-3,5
Keying


input
K
K
INV
NOR
+14V
330
E.F
E.F
EF
MIX
75Ω
75
75
PED
PEDCTL
VIDAMP
out
gate pulse
gate
Video Balance
gate Balance
-6V
K Gain
I
MV set
Clip
gate pulse AMP
SYNC
Burst

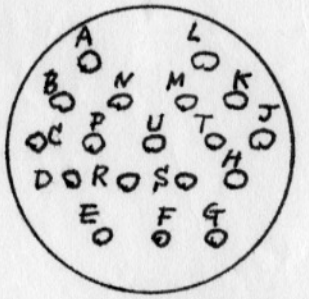

GND 1
VIDIN. 3
4
PED. CTL 5 6 7
CLP. 8 9
Croma IN 10
+15V 12
C. OUT 14
15
V. OUT 16
18
BLK. IN. 19
GND 22
2.2K
56K
3.3K
.1
47µ
2.2K
47K
10µ
10K
.005
130
10K
50µ
To. 8
5.6K
1.5K
50K
2.2K
270
1.2K
2.2K
10K
1K
A
B
CA-3019
330
30µ
5µh
50µ
270
5µ
22K
22K
75µ
1K
75
12K
33K
68K
To. 16
50
75
1.2K
1.5K
5.6K
100µ 6V
10K
820
2.2K
100µ
50µ
100µ
22K
56K
1.2K
5.6K
120
620
100µ
620
3.3K
PED. CONT.



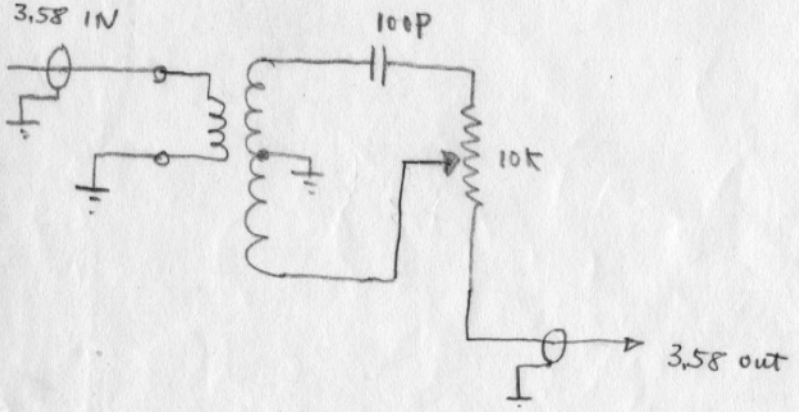
GATE Pulse Amp

CTC PANEL

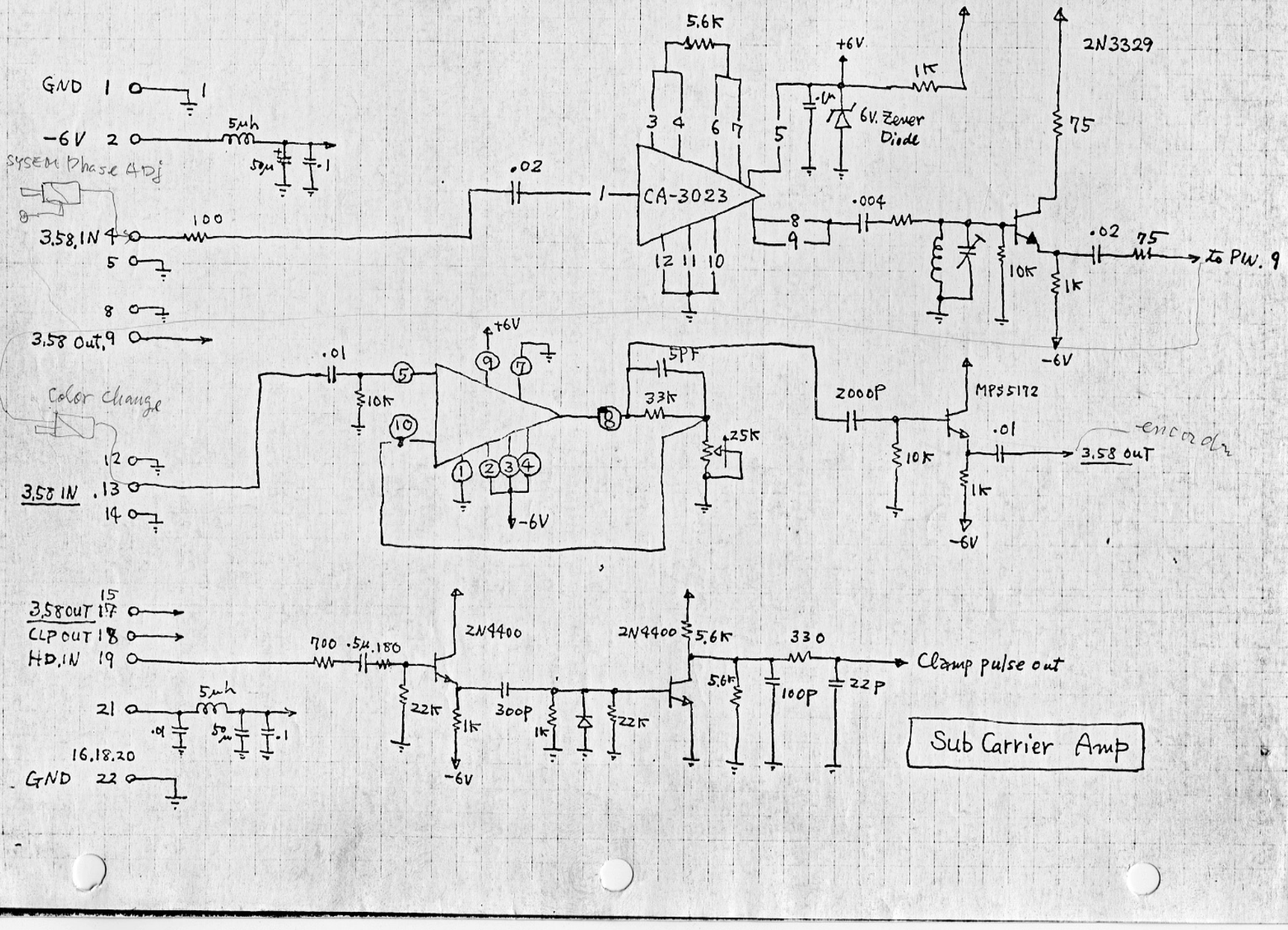
A	+ 12V
B	− 6V
P	GND
C	Rout
E	Coax SEL VID
H	F out
L	T out
F	GND
K	〃

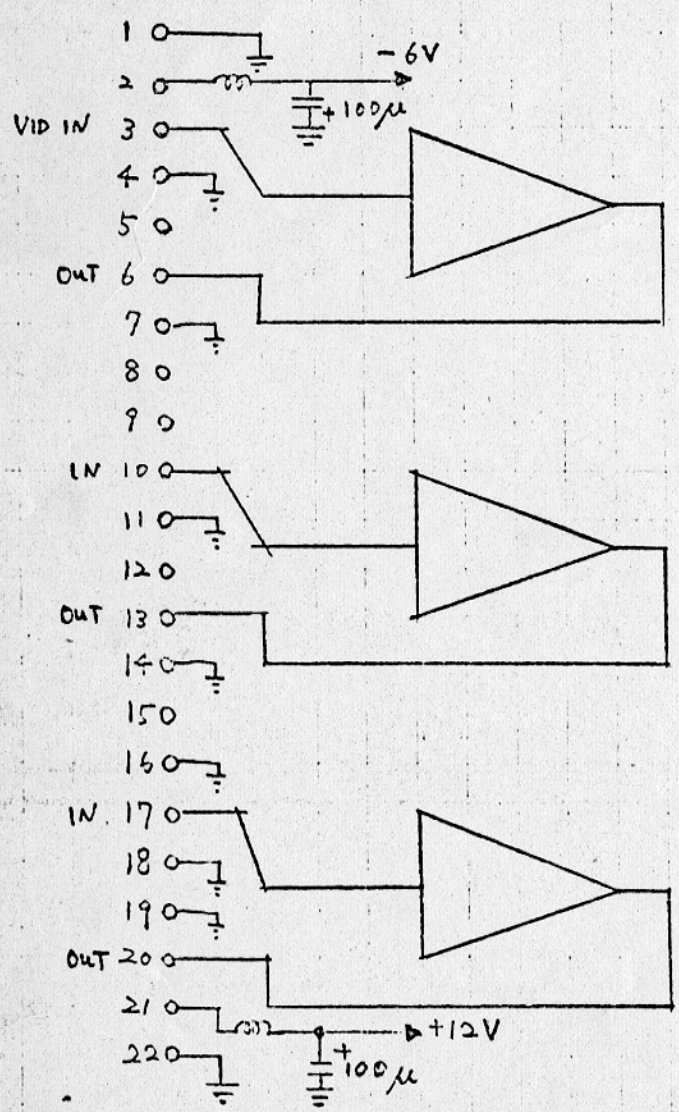


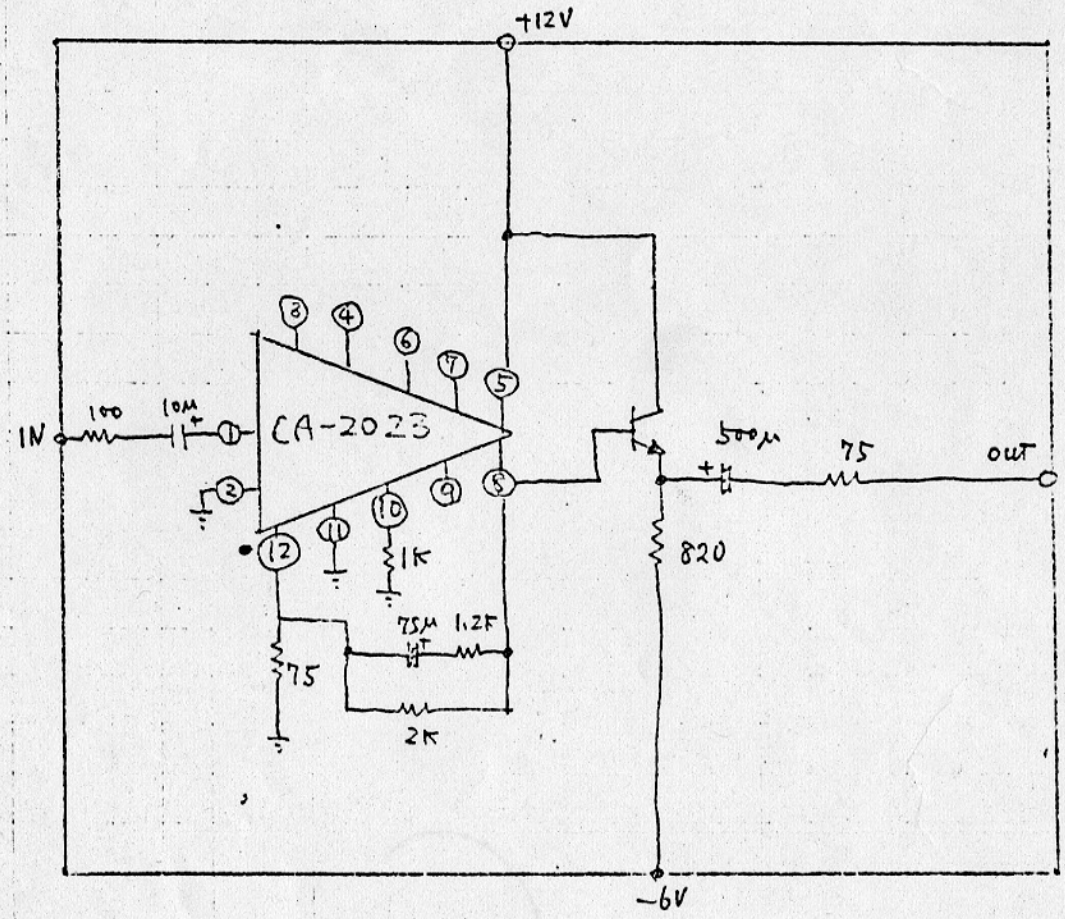
CONNETOR Diagram



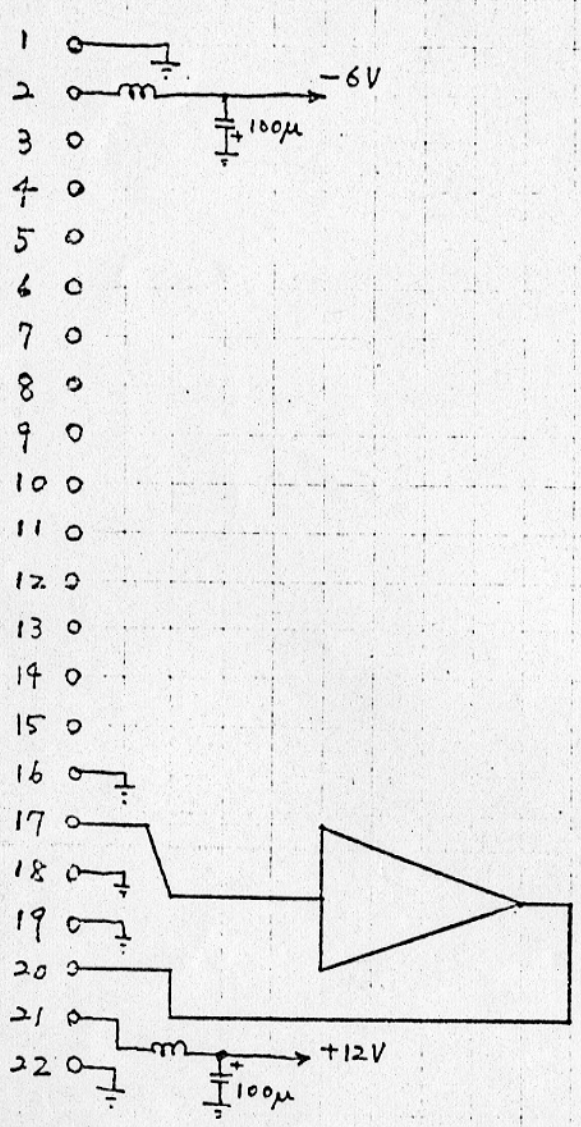
Phase CTL.

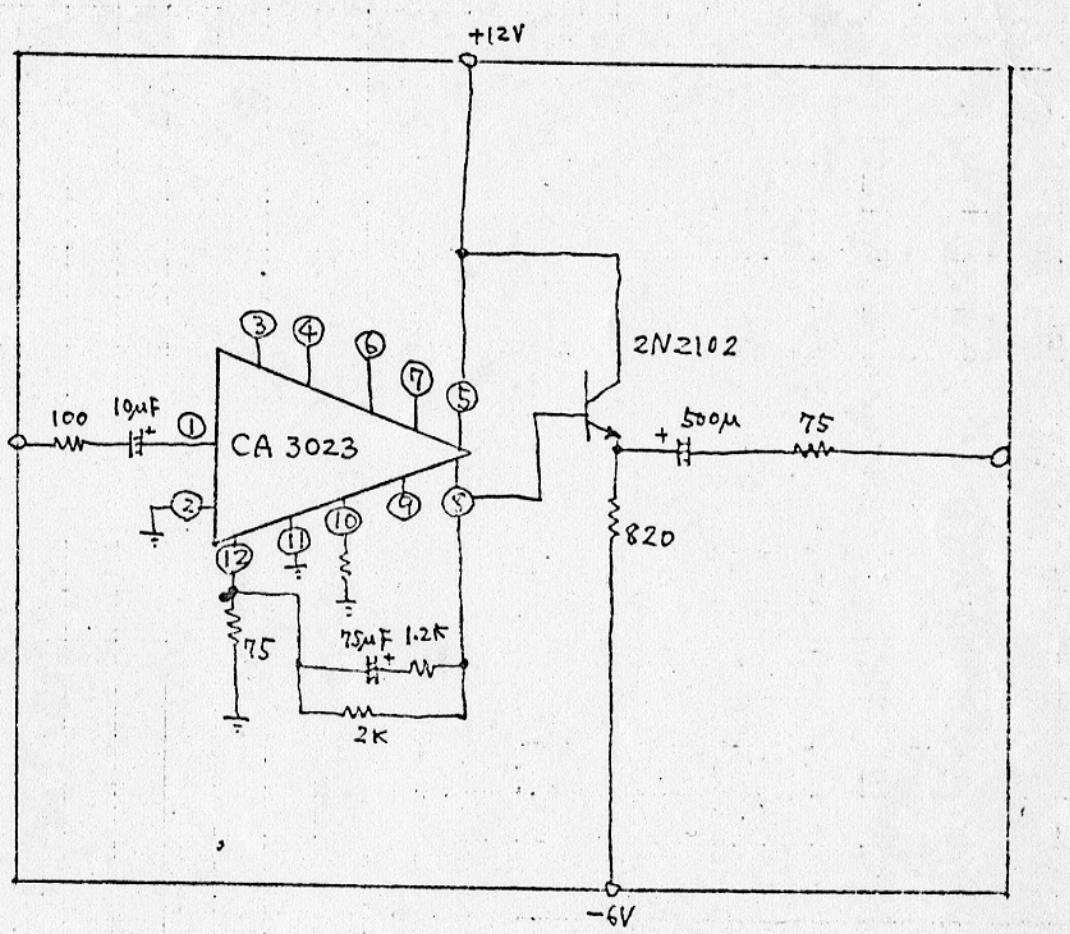

GND 1
-6V 2
5µh
50µ
.1
SYSEM Phase ADj
3.58 IN 4
100
5
8
3.58 Out 9
.02
CA-3023
3 4 6 7
5
12 11 10
5.6K
+6V
1K
.1µ
6V. Zener Diode
2N3329
75
.004
.02
75
to PIN. 9
10K
1K
-6V
Color Change
12
3.58 IN .13
14
.01
10K
+6V
5PF
33K
25K
2000P
MPS5172
.01
3.58 OUT
encoder
10K
1K
-6V
15
3.58OUT 17
CLP OUT 18
HD. IN 19
21
16.18.20
GND 22
700 .5µ .180
2N4400
22K
1K
-6V
300P
1K
22K
2N4400
5.6K
330
Clamp pulse out
5.6K
100P
22P
Sub Carrier Amp


1
2
VID IN 3
4
5
OUT 6
7
8
9
IN 10
11
12
OUT 13
14
15
16
IN 17
18
19
OUT 20
21
22
-6V
+100μ
+12V
+100μ

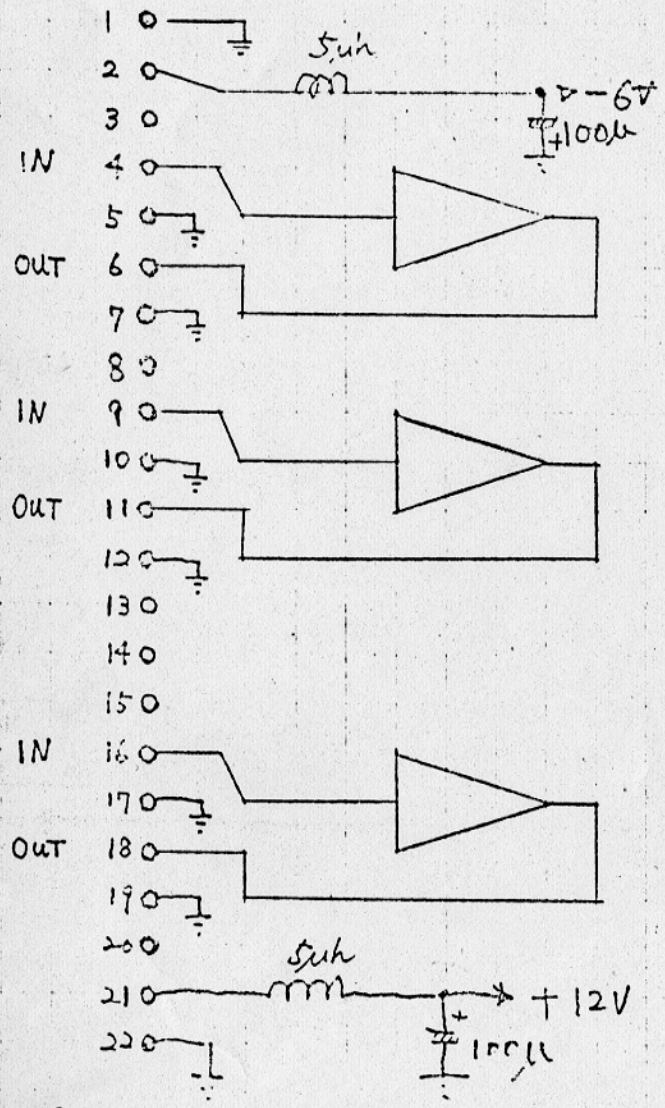

+12V
IN
100
10M
CA-2023
500μ
75
OUT
1K
820
75M
1.2K
75
2K
-6V

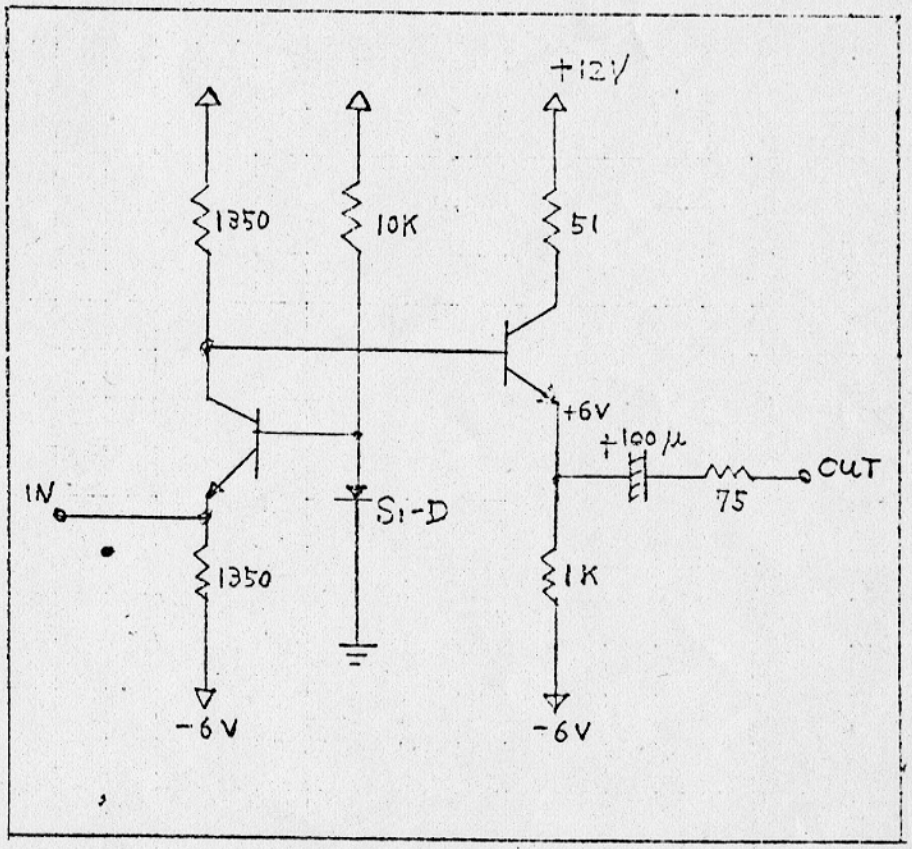
Amp.


1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
-6V
100μ
+12V
100μ

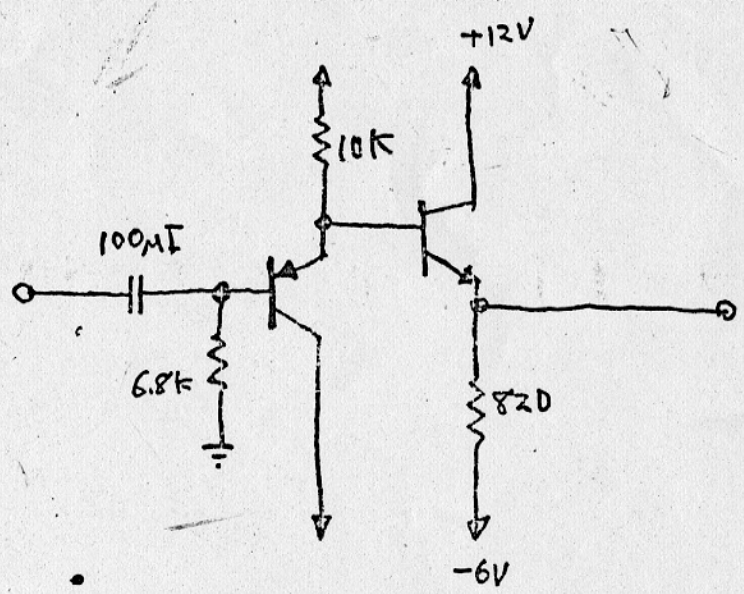

+12V
2N2102
100
10μF
CA 3023
500μ
75
820
75
75μF
1.2K
2K
-6V

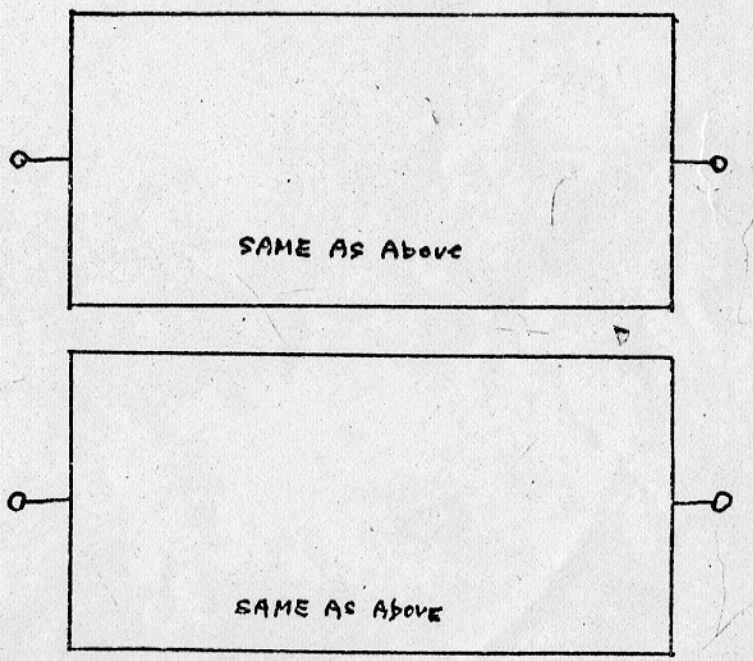
Amp.


1
2
3
IN 4
5
OUT 6
7
8
IN 9
10
OUT 11
12
13
14
15
IN 16
17
OUT 18
19
20
21
22
5μh
-6V
+100μ
5μh
+12V

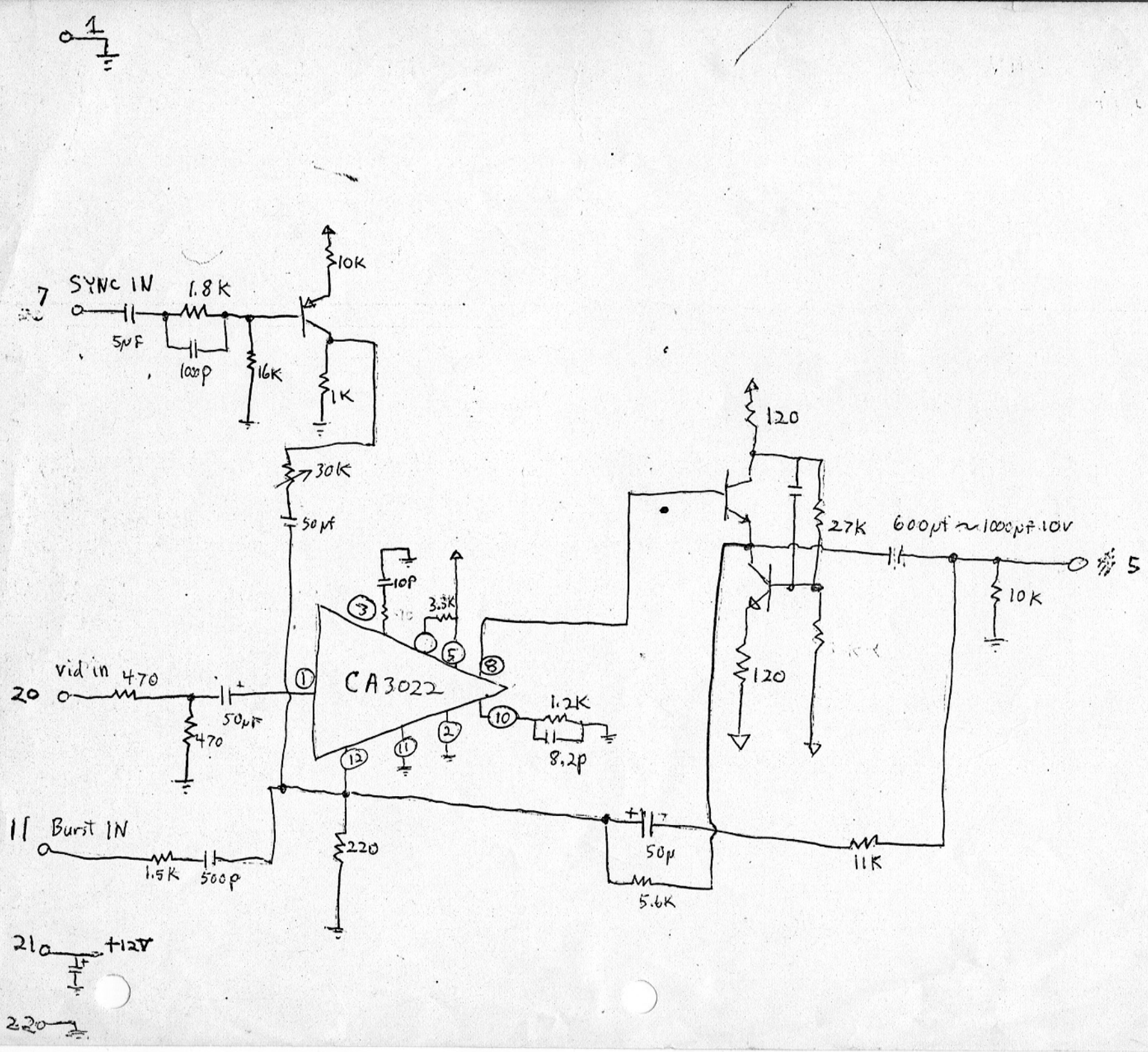

+12V
1350
10K
51
+6V
+100μ
OUT
75
IN
S1-D
1350
1K
-6V
-6V

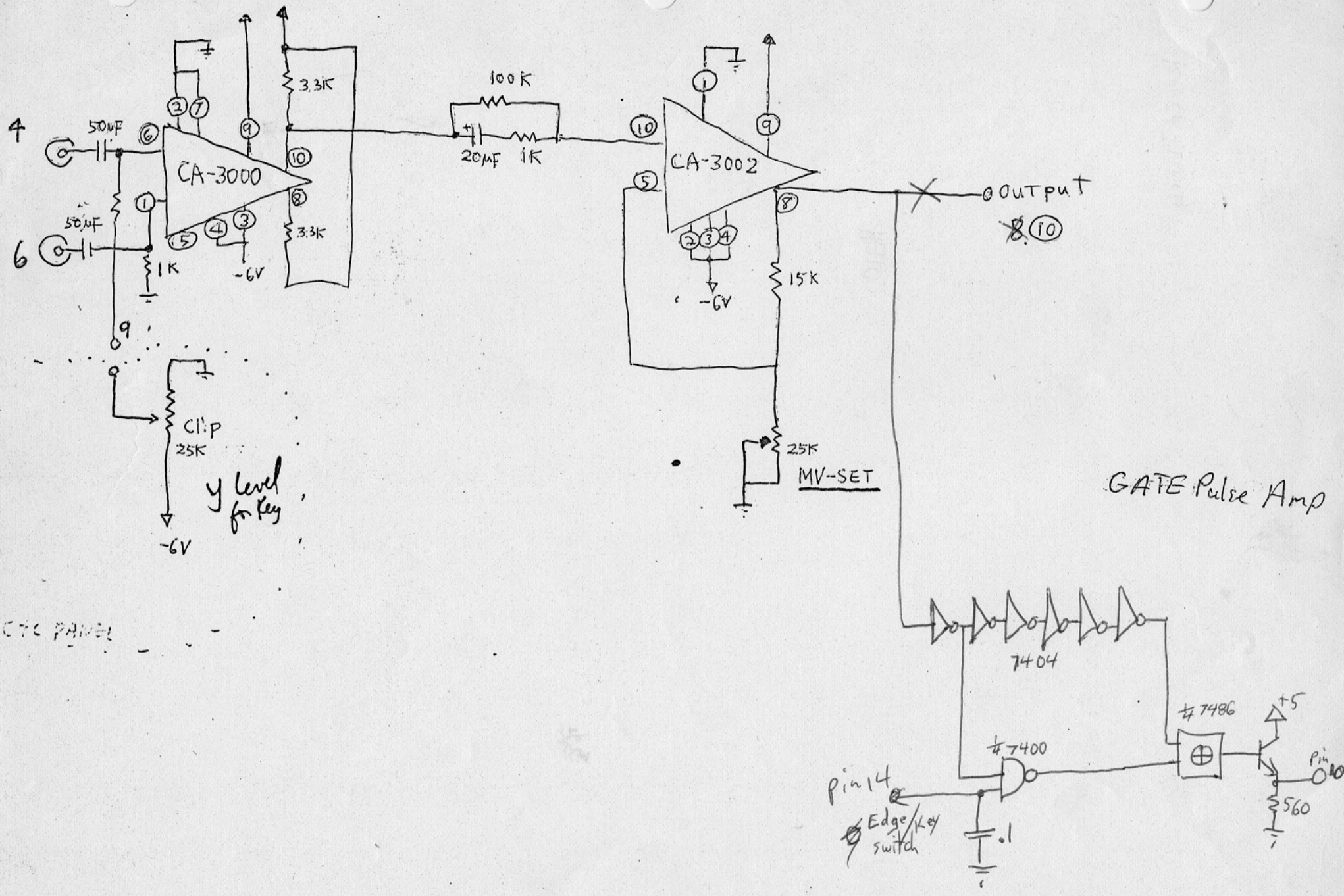
Mixing Amp

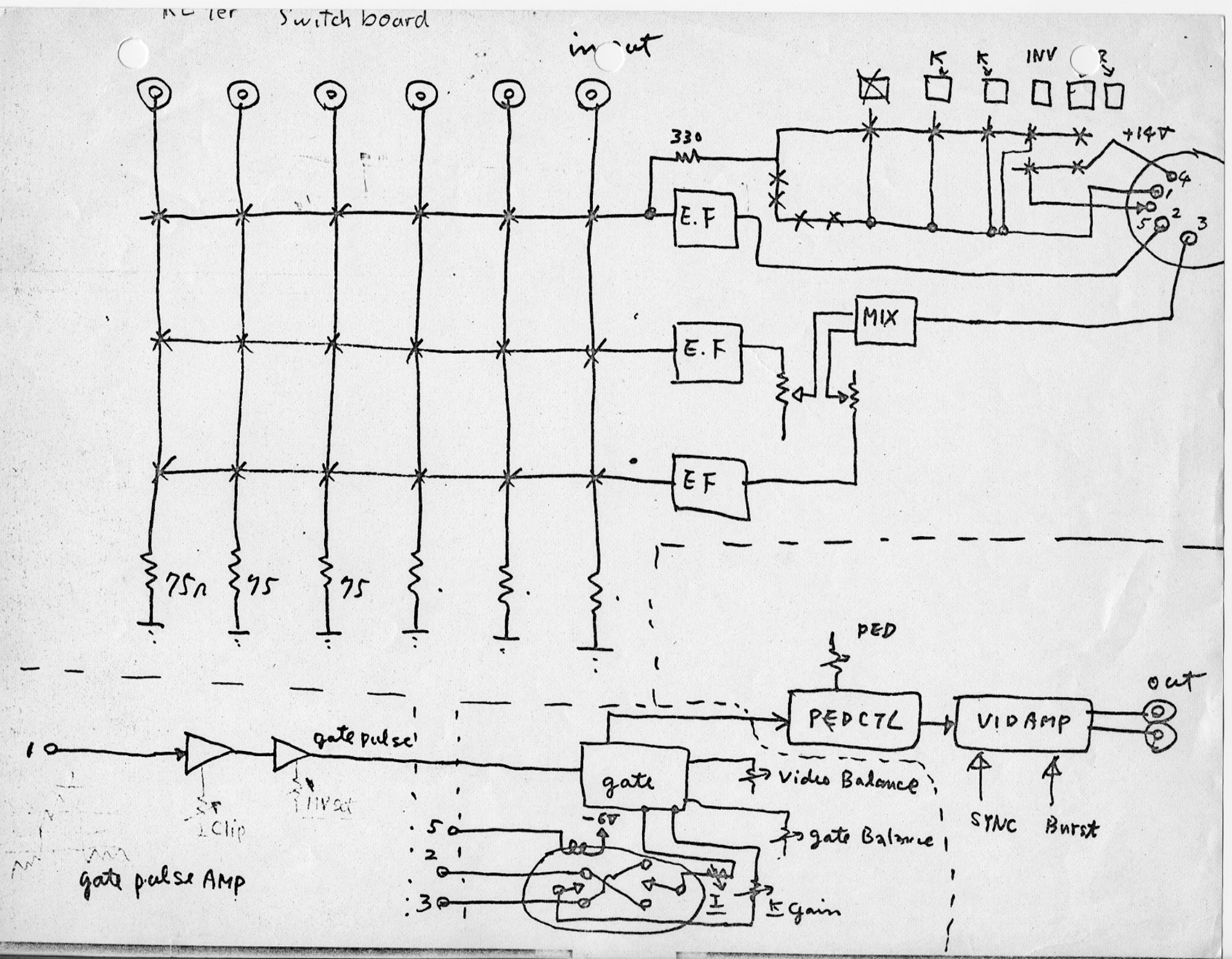


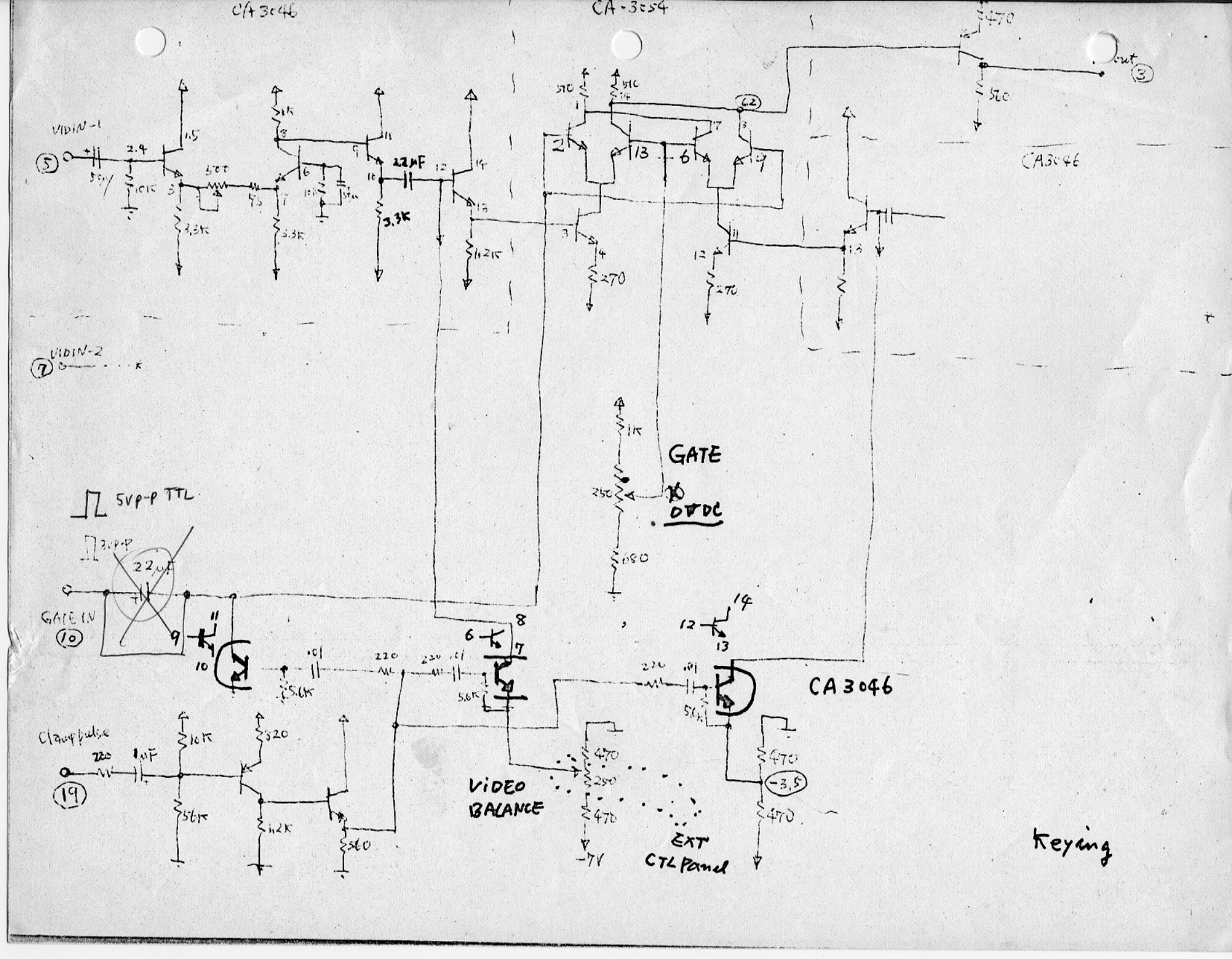


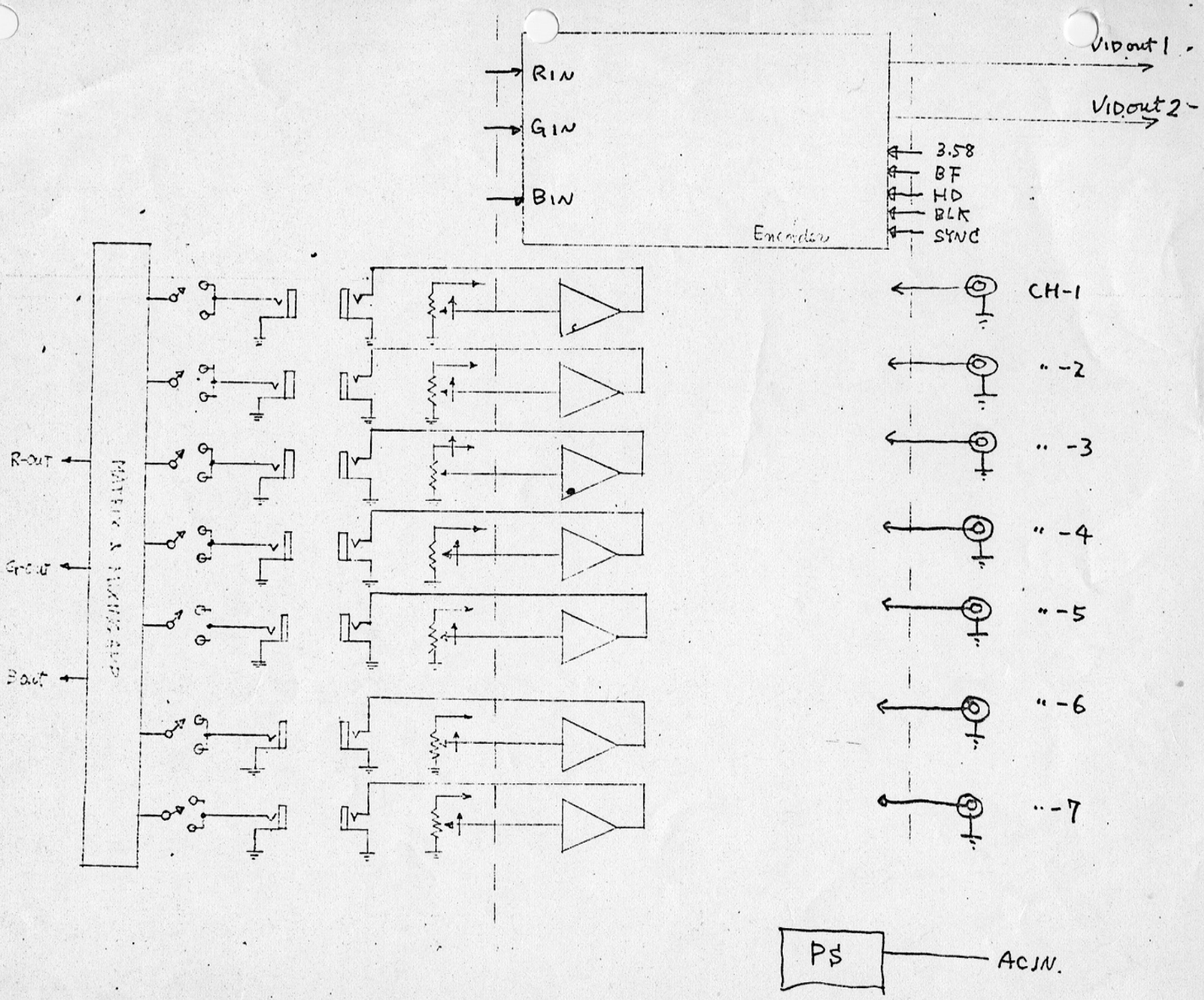
SWITCHER INPUT
EMITTER
Follower


1
7
SYNC IN
1.8K
5μF
1000p
16K
10K
1K
30K
50μf
10p
3.3K
3
5
8
1
CA3022
12
11
2
10
1.2K
8.2p
20
vid in
470
50μF
470
120
27K
600μf ~ 1000μF 10V
5
10K
120
11
Burst IN
1.5K
500p
220
50μ
5.6K
11K
21
+12V
220


4
50MF
6
50MF
1K
CA-3000
3.3K
3.3K
-6V
100K
20MF
1K
CA-3002
-6V
15K
25K
MV-SET
output
10
9
CLIP
25K
y level for key
-6V
GATE Pulse Amp
7404
#7486
#7400
+5
pin 14
Edge/Key switch
.1
560
Pin 10


Switch board
in out
INV
330
+14V
E.F
E.F
EF
MIX
75Ω
75
75
PED
PEDCTL
VIDAMP
out
gate pulse
gate
Video Balance
gate Balance
-6V
SYNC
Burst
Clip
gate pulse AMP
Gain


CA3046
CA-3054
CA3046
VIDIN-1
VIDIN-2
22MF
3.3K
GATE
0 V DC
5Vp-p TTL
GATE IN
Clamppulse
VIDEO BALANCE
EXT CTL Panel
CA3046
-7V
-3.5
Keying


RIN
GIN
BIN
Encoder
VIDout 1
VIDout 2
3.58
BF
HD
BLK
SYNC
CH-1
" -2
" -3
" -4
" -5
" -6
" -7
R-OUT
G-OUT
B-OUT
PS
ACIN.